

IMAT QUESTION BANK: Chemistry (Medium-Hard)

65 questions · 108 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · Question bank · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Organic chemistry and isomerism (Q1-13)

1. D-glucose and D-galactose are both aldohexoses with the same bonding order; their Fischer projections differ only at carbon 4, where the OH points right in glucose and left in galactose. Which term most precisely classifies the pair?

- A) They are enantiomers, because the OH at C-4 is on opposite sides in the two sugars.
- B) They are identical compounds drawn as different Fischer projections.
- C) They are epimers, because they differ in configuration at exactly one chiral center, C-4.
- D) They are constitutional isomers, because in galactose the C-4 OH is bonded to a different carbon.
- E) They are anomers, because any two aldohexoses that differ at a single carbon are anomers.

Answer: C

Both sugars share the same molecular formula and bonding order, and their Fischer projections differ at exactly one stereocenter, C-4. Sugars of identical connectivity that differ in configuration at a single chiral center are epimers by definition. The mirror-image claim fails because enantiomers require every stereocenter to be inverted, while C-2, C-3 and C-5 still match, and the OH is bonded to C-4 in both sugars, so no connectivity has changed.

2. A synthesis produces a molecule with the same bonding order as (2R,3S,4R)-hexane-2,3,4-triol but with configurations (2S,3R,4S) at its three chiral centers. Which term most precisely classifies the relationship between the two molecules?

- A) They are constitutional isomers, because the OH groups occupy different positions.
- B) They are epimers, because several chiral centers change at the same time.
- C) They are diastereomers, because some of the stereocenters are inverted.
- D) They are enantiomers, because every chiral center is inverted, giving two non-superimposable mirror images.
- E) They are identical molecules, because they share the same molecular formula.

Answer: D

The second molecule inverts all three stereocenters of (2R,3S,4R) to give (2S,3R,4S), so it is the exact mirror image of the first, and two non-superimposable mirror images are enantiomers. Diastereomers must keep at least one stereocenter unchanged, and epimers differ at exactly one, so neither label fits. The bonding order never changed, so they are not constitutional isomers either.

3. In water, D-glucose cyclises and exists as two forms that differ only at carbon 1: in the alpha form the new OH points down from the ring, in the beta form it points up, while the configurations at carbons 2 to 5 are identical. Which term most precisely classifies the two forms?

- A) Conformational isomers, interconvertible by rotation about a single bond.
- B) Anomers, because they differ in configuration at the anomeric carbon, C-1.
- C) Constitutional isomers, because the OH is attached to the ring in two different ways.
- D) Enantiomers, because the anomeric carbon is the most important stereocenter.
- E) Identical compounds, because the two forms have the same molecular formula.

Answer: B

The two cyclic forms differ only at C-1, the anomeric carbon created when the aldehyde closes into a hemiacetal ring, and sugars differing in configuration at the anomeric carbon are anomers by definition. They are not conformers, because switching alpha to beta requires breaking and re-forming the C-1 bond rather than rotating about it, and they are not mirror images because the configurations at C-2 to C-5 are identical.

4. Butane-2,3-diol exists in stereoisomeric forms: one has the configuration (2R,3R) and another has (2R,3S). Which term most precisely classifies these two forms?

- A) They are conformational isomers, because they interconvert by bond rotation at room temperature.
- B) They are constitutional isomers, because an OH group has moved from C-2 to C-3.
- C) They are enantiomers, because one form contains an R descriptor and the other an S descriptor.
- D) They are identical, because rotation about the C-2 to C-3 bond superimposes them.
- E) They are diastereomers, because they are stereoisomers that are not mirror images of each other.

Answer: E

The mirror image of (2R,3R) is (2S,3S), so the (2R,3S) form is not the mirror image of (2R,3R): the pair are stereoisomers that are not mirror images, which is the definition of diastereomers. Keeping one center unchanged at R is exactly what rules out the enantiomer description, and no bond rotation can interconvert them, because rotation about the C2-C3 bond never changes a configuration.

5. Compound X is $\text{CH}_3\text{OCH}_2\text{CH}_3$ (methoxyethane) and compound Y is $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (propan-1-ol); both have molecular formula $\text{C}_3\text{H}_8\text{O}$. Which statement is correct?

- A) They are constitutional isomers with different functional groups: an ether and an alcohol.
- B) They are enantiomers, because the alcohol is a chiral molecule.
- C) They are diastereomers, because their physical properties differ.
- D) They are the same compound drawn differently, since both have formula $\text{C}_3\text{H}_8\text{O}$.
- E) They are both ethers, because both contain a C-O single bond.

Answer: A

Both compounds have the formula $\text{C}_3\text{H}_8\text{O}$, but methoxyethane links its oxygen between two carbons (a C-O-C ether link) while propan-1-ol bonds its oxygen to hydrogen (a C-O-H alcohol group), so the atoms are connected differently: they are constitutional isomers of the functional-group type. The enantiomer claim fails because propan-1-ol has no chiral center, since its middle carbon carries two identical hydrogen atoms.

6. D-glucose is an aldohexose with an aldehyde group at C-1, while D-fructose is a ketohexose with a ketone group at C-2. Both have molecular formula $C_6H_{12}O_6$. Which term most precisely classifies the pair?

- A) Epimers, because they differ at one carbon atom.
- B) Stereoisomers, because only the position of the carbonyl differs.
- C) Constitutional isomers, because the same atoms are joined in a different order.
- D) Anomers, because the carbonyl sits at C-2 in one sugar and C-1 in the other.
- E) Enantiomers, because both are hexoses with the same formula.

Answer: C

Glucose and fructose share the formula $C_6H_{12}O_6$, but the carbonyl oxygen is joined to C-1 in the aldehyde and to C-2 in the ketone, so the same atoms are connected in a different order: they are constitutional isomers. Epimers and anomers must keep identical connectivity and differ only at a stereocenter, which is not the case, and moving a carbonyl is a connectivity change, not a stereochemical one.

7. The structure $CH_3CH_2CONH_2$ is named:

- A) Propanamide
- B) Propanenitrile
- C) Propanoic acid
- D) Propanamine
- E) Methyl ethanoate

Answer: A

A carbonyl carbon directly bonded to NH_2 is the amide group, and the three-carbon chain CH_3CH_2CO- gives the name propanamide. Propanamine would need an NH_2 attached to a saturated carbon with no $C=O$, methyl ethanoate is the ester CH_3COOCH_3 , and propanenitrile would carry a $C\equiv N$ group instead.

8. Oil of wintergreen (methyl 2-hydroxybenzoate) consists of a benzene ring bearing an OH group attached directly to the ring, plus a $COOCH_3$ group joined through the carbonyl carbon. Which two functional groups does the molecule contain?

- A) A phenol and an amide
- B) An ether and a ketone
- C) A carboxylic acid and an ester
- D) A phenol and an ester
- E) An alcohol and a carboxylic acid

Answer: D

An OH bonded directly to a benzene ring is a phenol rather than an alcohol, and the $COOCH_3$ linkage (carbonyl carbon joined through oxygen to CH_3) is an ester. No free $COOH$ remains, because the parent carboxylic acid group has been converted into the ester, so the molecule contains a phenol and an ester, not an alcohol or a carboxylic acid.

9. The structure $CH_3CH_2OCH_2CH_3$ is named:

- A) Butanal
- B) Ethoxyethane
- C) Butan-1-ol
- D) Butan-2-one
- E) Butanoic acid

Answer: B

One oxygen atom bridges two CH_2CH_3 groups, the defining C-O-C linkage of an ether, so the name is ethoxyethane, commonly called diethyl ether. Butan-1-ol shares the formula $C_4H_{10}O$ but has a terminal OH on a carbon chain, and butanal, butan-2-one and butanoic acid all require a carbonyl carbon, which this structure lacks.

10. Which of the following compounds is a ketone and contains exactly four carbon atoms?

- A)** Butanal, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
- B)** Propanone, CH_3COCH_3
- C)** Butan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$
- D)** Butanoic acid, $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
- E)** Butan-2-one, $\text{CH}_3\text{COCH}_2\text{CH}_3$

Answer: E

A ketone has its $\text{C}=\text{O}$ on an internal carbon of the chain. Butan-2-one, $\text{CH}_3\text{COCH}_2\text{CH}_3$, meets both requirements: a four-carbon chain and an internal carbonyl. Butanal and butanoic acid reach the carbon count but carry an aldehyde and a carboxylic acid group, propanone is a true ketone with only three carbons, and butan-2-ol is an alcohol with no $\text{C}=\text{O}$ at all.

11. An aldohexose is drawn as a Fischer projection with the aldehyde at the top as C-1. The OH groups at C-2, C-3 and C-4 point right, right and left respectively, and the OH at C-5, the chiral center farthest from the aldehyde, points left. How is the sugar classified?

- A)** It is an L-sugar, because the OH on the stereocenter farthest from the aldehyde points left.
- B)** It is meso, because left- and right-pointing OH groups balance two against two.
- C)** It is a D-sugar, because the aldehyde group is drawn at the top of the projection.
- D)** It is a D-sugar, because D/L is fixed by the OH closest to the aldehyde group.
- E)** It is a D-sugar, because two of its four OH groups point right.

Answer: A

The D or L label is decided solely by the chiral center farthest from the carbonyl group, C-5 in an aldohexose: a left-pointing OH there places the sugar in the L series. What happens at C-2 to C-4, including any two-right, two-left balance of the OH groups, is irrelevant to the label, and meso describes internal mirror symmetry, not the D/L classification.

12. Maltose and cellobiose are both disaccharides of two glucose units with molecular formula $\text{C}_{12}\text{H}_{22}\text{O}_{11}$. Maltose joins its units with an α -1,4 glycosidic bond and cellobiose with a β -1,4 bond. Human enzymes hydrolyse maltose efficiently but cannot digest cellobiose. Which statement explains this best?

- A)** Maltose is digested because it is a reducing sugar, whereas cellobiose is non-reducing.
- B)** Cellobiose passes through the gut unchanged because no enzyme can release glucose from a β bond.
- C)** Cellobiose cannot be digested because its molecular formula differs from that of maltose.
- D)** Human enzymes fit the α -1,4 geometry of maltose but not the β -1,4 geometry of cellobiose, so cellobiose is not hydrolysed.
- E)** Cellobiose is indigestible because β glycosidic bonds are chemically stronger than α bonds and cannot be broken in water.

Answer: D

The two disaccharides contain exactly the same atoms; only the three-dimensional geometry of the link differs, α -1,4 in maltose versus β -1,4 in cellobiose. Enzyme active sites are stereochemically specific, so the human enzyme fits the α geometry and hydrolyses maltose but cannot bind cellobiose. The claim that β bonds are chemically stronger fails because both bond types hydrolyse given the right catalyst (human lactase even digests the β bond of lactose), and cellobiose, with one free anomeric carbon, is itself a reducing sugar.

13. In sucrose, the glycosidic bond links the anomeric carbon C-1 of glucose to the anomeric carbon C-2 of fructose. In maltose, the glycosidic bond uses the anomeric carbon of only one of its two glucose units. Which statement is correct?

- A)** Sucrose is reducing because fructose is a ketose, and ketoses are always stronger reducing agents.
- B)** Sucrose is non-reducing because both anomeric carbons are locked inside its glycosidic bond, while maltose is reducing because one anomeric carbon stays free.
- C)** Maltose is non-reducing, because it is built only from glucose units.
- D)** Sucrose is reducing, because the fructose ring can still open at C-2 while glycosidically bonded.
- E)** Both sugars are non-reducing, because any glycosidic bond blocks reduction.

Answer: B

A sugar is reducing only if it has a free anomeric carbon that can open to the carbonyl form. Sucrose locks both anomeric carbons (C-1 of glucose and C-2 of fructose) inside its glycosidic bond, so neither ring can open and sucrose is non-reducing, while maltose leaves the second glucose's anomeric carbon free and is reducing. The claim that any glycosidic bond blocks reduction fails because only the anomeric centers taking part in the bond are locked, not every unit in the molecule.

Section B - Stoichiometry, moles and gas laws (Q14-27)

14. How many molecules are present in 4.4 g of propane, C_3H_8 ? ($H = 1$, $C = 12$; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

- A)** 1.204×10^{23} **B)** 3.01×10^{22} **C)** 6.02×10^{21} **D)** 6.02×10^{22} **E)** 6.02×10^{23}

Answer: D

Step 1: $M(C_3H_8) = 3(12) + 8(1) = 44 \text{ g/mol}$

Step 2: $n = 4.4 \div 44 = 0.10 \text{ mol}$

Step 3: molecules = $0.10 \times 6.02 \times 10^{23} = 6.02 \times 10^{22}$

The 6.02×10^{23} answer treats 4.4 g as a whole mole, and 1.204×10^{23} doubles the correct count.

15. How many hydrogen atoms are present in 0.25 mol of water, H_2O ? (Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

- A)** 1.204×10^{24} **B)** 1.505×10^{23} **C)** 3.01×10^{23} **D)** 6.02×10^{23} **E)** 9.03×10^{23}

Answer: C

Step 1: each H_2O carries two H atoms, so $n(H) = 0.25 \times 2 = 0.50 \text{ mol}$.

Step 2: H atoms = $0.50 \times 6.02 \times 10^{23} = 3.01 \times 10^{23}$

The 1.505×10^{23} value counts water molecules and drops the factor of two, and 9.03×10^{23} counts all three atoms per molecule instead of the hydrogens alone.

16. What mass of ammonia, NH_3 , contains the same number of molecules as 4.4 g of carbon dioxide, CO_2 ? ($C = 12$, $O = 16$, $N = 14$, $H = 1$)

- A)** 1.7 g **B)** 3.4 g **C)** 4.4 g **D)** 8.5 g **E)** 17 g

Answer: A

Step 1: $n(CO_2) = 4.4 \div 44 = 0.10 \text{ mol}$

Step 2: equal numbers of molecules mean equal moles, so $n(NH_3) = 0.10 \text{ mol}$.

Step 3: $M(NH_3) = 14 + 3(1) = 17 \text{ g/mol}$, so mass = $0.10 \times 17 = 1.7 \text{ g}$

Equal masses would contain equal molecule numbers only if the molar masses matched, which they do not here.

17. What volume does 8.8 g of carbon dioxide, CO_2 , occupy at STP? (C = 12, O = 16; molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$)

- A)** 4.48 dm^3 **B)** 8.8 dm^3 **C)** 11.2 dm^3 **D)** 22.4 dm^3 **E)** 44.8 dm^3

Answer: A

Step 1: $n(\text{CO}_2) = 8.8 \div 44 = 0.20 \text{ mol}$

Step 2: $V = 0.20 \times 22.4 = 4.48 \text{ dm}^3$ at STP

The 11.2 dm^3 answer corresponds to 0.5 mol and the 22.4 dm^3 answer to 1 mol, neither of which matches 8.8 g of carbon dioxide.

18. How many molecules are present in 560 cm^3 of an ideal gas at STP? ($1 \text{ dm}^3 = 1000 \text{ cm}^3$; molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

- A)** 1.505×10^{22} **B)** 1.505×10^{24} **C)** 1.505×10^{25} **D)** 6.02×10^{21} **E)** 6.02×10^{23}

Answer: A

Step 1: $560 \text{ cm}^3 = 0.560 \text{ dm}^3$

Step 2: $n = 0.560 \div 22.4 = 0.025 \text{ mol}$

Step 3: molecules = $0.025 \times 6.02 \times 10^{23} = 1.505 \times 10^{22}$

The 1.505×10^{25} answer skips the cm^3 to dm^3 conversion and divides 560 by 22.4 as though the volume were already in cubic decimetres.

19. What is the mass of 4.48 dm^3 of oxygen gas, O_2 , measured at STP? (O = 16; molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$)

- A)** 1.6 g **B)** 3.2 g **C)** 6.4 g **D)** 16 g **E)** 32 g

Answer: C

Step 1: $n(\text{O}_2) = 4.48 \div 22.4 = 0.20 \text{ mol}$

Step 2: $M(\text{O}_2) = 2 \times 16 = 32 \text{ g/mol}$

Step 3: mass = $0.20 \times 32 = 6.4 \text{ g}$

The 3.2 g answer forgets that oxygen gas is diatomic and uses the atomic mass 16 rather than the molecular mass 32.

20. 10.0 g of calcium carbonate ($M = 100 \text{ g/mol}$) is treated with 100 cm^3 of 1.0 mol/dm^3 hydrochloric acid: $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$. What volume of CO_2 is released, measured at STP? ($1 \text{ dm}^3 = 1000 \text{ cm}^3$; molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$)

- A)** 0.56 dm^3 **B)** 1.12 dm^3 **C)** 2.24 dm^3 **D)** 11.2 dm^3 **E)** 22.4 dm^3

Answer: B

Step 1: $n(\text{CaCO}_3) = 10.0 \div 100 = 0.10 \text{ mol}$, which would consume 0.20 mol HCl at the 1:2 ratio. Step 2: $n(\text{HCl}) = 0.100 \text{ dm}^3 \times 1.0 \text{ mol/dm}^3 = 0.10 \text{ mol}$, less than needed, so the acid runs out first.

Step 3: $n(\text{CO}_2) = 0.10 \div 2 = 0.05 \text{ mol}$

Step 4: $V = 0.05 \times 22.4 = 1.12 \text{ dm}^3$

The 2.24 dm^3 answer treats the carbonate as limiting and ignores the 2 mol of HCl required per mol of CO_2 .

21. In theory, 9.2 g of ethanol ($M = 46 \text{ g/mol}$) dehydrates completely to ethene ($M = 28 \text{ g/mol}$): $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}$. The actual mass of ethene collected is 4.2 g. What is the percentage yield?

- A)** 25% **B)** 56% **C)** 75% **D)** 100% **E)** 133%

Answer: C

Step 1: $n(\text{ethanol}) = 9.2 \div 46 = 0.20 \text{ mol}$, so at most 0.20 mol of ethene can form.

Step 2: theoretical mass $= 0.20 \times 28 = 5.6 \text{ g}$

Step 3: percentage yield $= (4.2 \div 5.6) \times 100 = 75\%$

The 25% figure reports the 1.4 g shortfall as a share of the theoretical mass instead of the fraction obtained, and 133% inverts the ratio.

22. A diver exhales an air bubble of volume 2.0 dm^3 at a depth where the pressure is 400 kPa. The temperature stays constant as the bubble rises to the surface, where the pressure is 100 kPa. What is the volume of the bubble at the surface?

- A)** 0.5 dm^3 **B)** 1.0 dm^3 **C)** 2.0 dm^3 **D)** 4.0 dm^3 **E)** 8.0 dm^3

Answer: E

At constant temperature, Boyle's law gives $P_1V_1 = P_2V_2$, so $V_2 = (400 \times 2.0) \div 100 = 8.0 \text{ dm}^3$: a fourfold pressure drop forces a fourfold expansion. The 0.5 dm^3 answer inverts the pressure ratio and wrongly predicts that the bubble shrinks as it rises toward lower pressure.

23. A rigid sealed canister holds a fixed amount of gas at 27°C and 100 kPa. It is heated until the gas reaches 327°C . What is the new pressure inside the canister?

- A)** 50 kPa **B)** 100 kPa **C)** 109 kPa **D)** 200 kPa **E)** 300 kPa

Answer: D

Step 1: convert to kelvin, $27^\circ\text{C} = 300 \text{ K}$ and $327^\circ\text{C} = 600 \text{ K}$

Step 2: with volume and moles fixed, $P_2 = P_1 \times (T_2 \div T_1) = 100 \times (600 \div 300) = 200 \text{ kPa}$. The 109 kPa answer comes from multiplying 100 by $327 \div 300$, using Celsius readings in a ratio that only kelvin values may enter.

24. A gas occupies 9.0 dm^3 at 27°C and 100 kPa. What volume does it occupy at 127°C and 200 kPa?

- A)** 4.5 dm^3 **B)** 6.0 dm^3 **C)** 12 dm^3 **D)** 13.5 dm^3 **E)** 24 dm^3

Answer: B

The combined gas law gives $V_2 = V_1 \times (T_2 \div T_1) \times (P_1 \div P_2) = 9.0 \times (400 \div 300) \times (100 \div 200) = 6.0 \text{ dm}^3$, using 300 K and 400 K. Warming expands the gas but the doubled pressure compresses it more, so the net effect is a shrink from 9.0 to 6.0 dm^3 . The 12 dm^3 answer ignores the pressure change entirely, and 13.5 dm^3 inverts both ratios.

25. A sealed vessel contains 0.20 mol of nitrogen gas and 0.30 mol of oxygen gas at constant temperature, and the total pressure is 150 kPa. What is the partial pressure of the oxygen?

- A)** 45 kPa **B)** 60 kPa **C)** 75 kPa **D)** 90 kPa **E)** 150 kPa

Answer: D

Step 1: total moles $= 0.20 + 0.30 = 0.50 \text{ mol}$

Step 2: mole fraction of $\text{O}_2 = 0.30 \div 0.50 = 0.60$

Step 3: $p(\text{O}_2) = 0.60 \times 150 = 90 \text{ kPa}$

The 60 kPa value is the nitrogen share, and 45 kPa multiplies the raw 0.30 mol by the total pressure without first converting to a mole fraction.

26. 100 cm³ of 0.80 mol/dm³ hydrochloric acid is mixed with 400 cm³ of 0.20 mol/dm³ hydrochloric acid. Assuming the volumes are additive, what is the concentration of the resulting solution?

- A)** 0.16 mol/dm³ **B)** 0.20 mol/dm³ **C)** 0.32 mol/dm³ **D)** 0.50 mol/dm³ **E)** 0.80 mol/dm³

Answer: C

Step 1: moles from the concentrated acid = $0.100 \times 0.80 = 0.080$ mol. Step 2: moles from the dilute acid = $0.400 \times 0.20 = 0.080$ mol.

Step 3: total = 0.160 mol in 0.500 dm³, so $c = 0.160 \div 0.500 = 0.32$ mol/dm³

The 0.50 mol/dm³ answer takes a plain average of the two concentrations, ignoring that four times more of the dilute solution was used.

27. A solution is prepared by diluting 25 cm³ of a 2.0 mol/dm³ solution to a concentration of 0.10 mol/dm³. What is the final volume of the diluted solution?

- A)** 25 cm³ **B)** 250 cm³ **C)** 450 cm³ **D)** 500 cm³ **E)** 1250 cm³

Answer: D

Step 1: moles of solute = $0.025 \text{ dm}^3 \times 2.0 \text{ mol/dm}^3 = 0.050$ mol

Step 2: final volume = $0.050 \div 0.10 = 0.50 \text{ dm}^3 = 500 \text{ cm}^3$

The question asks for the final volume of the diluted solution, so the original 25 cm³ must not be subtracted; that subtraction would answer a water-added question instead.

Section C - Solutions, acids and bases (Q28-40)

28. What is the pH of a 2.0×10^{-4} mol/dm³ solution of hydrochloric acid, a strong acid that is fully ionised in water?

- A)** 2.7 **B)** 3.3 **C)** 3.7 **D)** 4.0 **E)** 10.3

Answer: C

HCl is fully ionised, so $[\text{H}^+] = 2.0 \times 10^{-4} \text{ mol/dm}^3$ and $\text{pH} = -\log(2.0 \times 10^{-4}) = 4 - \log 2 = 4 - 0.30 = 3.70$. The 4.0 answer drops the 0.30 contribution of the factor 2, and the 10.3 value would describe a basic solution, impossible for a strong acid.

29. What is the pOH of a 5.0×10^{-2} mol/dm³ solution of nitric acid, a strong acid that is fully ionised in water? ($K_w = 1.0 \times 10^{-14}$ at 25 °C)

- A)** 1.3 **B)** 2.7 **C)** 12.3 **D)** 12.7 **E)** 13.7

Answer: D

Step 1: $[\text{H}^+] = 5.0 \times 10^{-2} \text{ mol/dm}^3$, so $\text{pH} = 2 - \log 5 = 2 - 0.70 = 1.30$

Step 2: $\text{pOH} = 14.00 - 1.30 = 12.70$

The 1.3 answer reports the pH, but pOH is the quantity asked for, obtained through K_w .

30. Strontium hydroxide, $\text{Sr}(\text{OH})_2$, is a strong base that is fully dissociated in water. What is the pH of a 5.0×10^{-4} mol/dm³ solution of $\text{Sr}(\text{OH})_2$? ($K_w = 1.0 \times 10^{-14}$ at 25 °C)

- A)** 3.0 **B)** 10.7 **C)** 11.0 **D)** 11.7 **E)** 12.0

Answer: C

Step 1: each $\text{Sr}(\text{OH})_2$ releases two OH^- ions, so $[\text{OH}^-] = 2 \times 5.0 \times 10^{-4} = 1.0 \times 10^{-3} \text{ mol/dm}^3$.

Step 2: $\text{pOH} = 3.00$ and $\text{pH} = 14.00 - 3.00 = 11.0$

The 10.7 answer forgets the factor of two and keeps $\text{pOH} = 3.3$, and the 3.0 value quotes the pOH as though it were the pH of a base.

31. Four solutions are prepared at 25 °C ($K_w = 1.0 \times 10^{-14}$) from fully ionised solutes: P is 1.0×10^{-3} mol/dm³ HCl, Q is 5.0×10^{-3} mol/dm³ HCl, R is 1.0×10^{-3} mol/dm³ NaOH and S is 1.0×10^{-3} mol/dm³ Ba(OH)₂. Which option places the four solutions in order of increasing pH?

- A)** Q, P, S, R **B)** P, Q, R, S **C)** S, R, Q, P **D)** P, Q, S, R **E)** Q, P, R, S

Answer: E

Step 1: P (10^{-3} M HCl) has pH 3.0; Q (5×10^{-3} M HCl) has pH $3 - 0.70 = 2.3$, so the more concentrated acid is the more acidic. Step 2: R (10^{-3} M NaOH) gives pOH 3.0 and pH 11.0. Step 3: S (10^{-3} M Ba(OH)₂) releases two OH⁻ per formula unit, so $[\text{OH}^-] = 2 \times 10^{-3}$, pOH = $3 - 0.30 = 2.7$ and pH 11.3. Increasing pH therefore runs Q, P, R, S; ordering by concentration alone (which puts P ahead of Q) ignores what the logarithm does to concentration.

32. Pure water is heated from 25 °C to 60 °C. As the temperature rises, K_w increases, and the pH of pure water at 60 °C is measured to be 6.5. Which of the following statements is correct?

- A)** The water is basic, because heating pushes the hydroxide concentration above the hydrogen ion concentration.
B) The water is acidic, because a pH below 7 always means more H₃O⁺ than OH⁻.
C) The water is neutral, because K_w keeps the value 1.0×10^{-14} at every temperature.
D) The water is acidic, because heating destroys hydroxide ions faster than hydrogen ions.
E) The water is neutral, because $[\text{H}^+]$ and $[\text{OH}^-]$ stay equal while both become larger than at 25 °C.

Answer: E

Pure water always produces H₃O⁺ and OH⁻ in equal amounts, so it remains neutral at every temperature; neutrality means equal concentrations of the two ions, not a pH of exactly 7, which is a 25 °C coincidence tied to $K_w = 1.0 \times 10^{-14}$. Heating raises K_w , so both concentrations climb above 10^{-7} mol/dm³ and the pH falls to 6.5 without any excess of H₃O⁺ over OH⁻.

33. At 10 °C the ionic product of water has fallen to $K_w = 3.0 \times 10^{-15}$ mol² dm⁻⁶. What is the hydroxide ion concentration, $[\text{OH}^-]$, in pure water at 10 °C?

- A)** 1.0×10^{-7} mol/dm³
B) 1.7×10^{-8} mol/dm³
C) 3.0×10^{-15} mol/dm³
D) 5.5×10^{-8} mol/dm³
E) 9.0×10^{-8} mol/dm³

Answer: D

In pure water $[\text{H}^+] = [\text{OH}^-]$, so $K_w = [\text{OH}^-]^2$ and $[\text{OH}^-] = \sqrt{(3.0 \times 10^{-15})} \approx 5.5 \times 10^{-8}$ mol/dm³. The 1.0×10^{-7} value keeps the 25 °C concentration although K_w has fallen, and 3.0×10^{-15} mistakes the ionic product itself for a concentration.

34. 10 cm³ of 0.10 mol/dm³ hydrochloric acid, fully ionised, is diluted with water to a final volume of 1.0 dm³. By how much does the pH of the solution change?

- A)** It stays the same, because the amount of acid is unchanged.
B) It reaches pH 7, because a hundredfold dilution turns any strong acid neutral.
C) It falls by 2 units.
D) It rises by 1 unit.
E) It rises by 2 units.

Answer: E

Step 1: initial $[\text{H}^+] = 0.10$ mol/dm³, pH 1.0

Step 2: the amount of HCl (1.0×10^{-3} mol) is unchanged, but the volume grows a hundredfold to 1.0 dm³, so $[\text{H}^+] = 1.0 \times 10^{-3}$ mol/dm³ and pH 3.0. The pH rises by exactly 2 units. Dilution is not neutralisation: a 10^{-3} mol/dm³ strong acid still has pH 3, not 7.

35. What volume of water must be added to 20 cm³ of 1.0 × 10⁻² mol/dm³ hydrochloric acid, fully ionised, to obtain a solution of pH 4.0 at 25 °C?

- A)** 180 cm³ **B)** 198 cm³ **C)** 200 cm³ **D)** 1980 cm³ **E)** 2000 cm³

Answer: D

Step 1: pH 4.0 requires [H⁺] = 1.0 × 10⁻⁴ mol/dm³, and the stock supplies 1.0 × 10⁻² mol/dm³, so a hundredfold dilution is needed.

Step 2: final volume = 20 × 100 = 2000 cm³

Step 3: water to add = 2000 – 20 = 1980 cm³

The 2000 cm³ answer forgets that the 20 cm³ of stock already occupies part of the final volume.

36. Which of the following mixtures, made up with pure water as solvent, can act as a buffer solution?

- A)** 100 cm³ of 0.10 mol/dm³ ethanoic acid mixed with 100 cm³ of 0.10 mol/dm³ sodium hydroxide.
B) 100 cm³ of 0.10 mol/dm³ ethanoic acid mixed with 50 cm³ of 0.10 mol/dm³ sodium hydroxide.
C) 100 cm³ of 0.10 mol/dm³ hydrochloric acid mixed with 50 cm³ of 0.10 mol/dm³ sodium hydroxide.
D) A 0.10 mol/dm³ solution of ethanoic acid alone in water.
E) 100 cm³ of 0.10 mol/dm³ hydrochloric acid mixed with 100 cm³ of 0.10 mol/dm³ sodium chloride.

Answer: B

A buffer requires comparable amounts of a weak acid and its conjugate base. Mixing 100 cm³ of 0.10 mol/dm³ ethanoic acid with 50 cm³ of 0.10 mol/dm³ NaOH neutralises exactly half the acid, leaving 0.005 mol of CH₃COOH and 0.005 mol of CH₃COO⁻ side by side in solution. Exact neutralisation leaves only sodium ethanoate with no acid reserve, either strong-acid mixture leaves excess HCl or mere spectator ions, and a weak acid alone carries essentially no conjugate base to absorb added acid.

37. A buffer solution contains comparable amounts of ethanoic acid and sodium ethanoate and has pH 4.8. An equal volume of pure water is added to it. Which of the following statements is correct?

- A)** The pH rises by about one unit because both concentrations fall by a factor of two.
B) The pH falls because the acid becomes effectively stronger on dilution.
C) The buffer is destroyed, because dilution pushes the dissociation of the acid completely towards the ions.
D) The pH and the buffer capacity are both unchanged, because a buffer resists any change made to it.
E) The pH stays close to 4.8 because the ratio of ethanoate to ethanoic acid is unchanged, but the buffer capacity falls.

Answer: E

The pH of an ethanoic acid and ethanoate buffer is governed by the ratio [CH₃COO⁻] ÷ [CH₃COOH]. Adding an equal volume of water divides both concentrations by the same factor, so the ratio, and hence the pH near 4.8, stays essentially unchanged, while the absolute amounts of acid and base available to absorb any added acid or alkali shrink. Dilution costs a buffer its capacity, not its pH, so the claims that the buffer is destroyed or that the pH jumps by about a unit misread which property the ratio controls.

38. Which of the following statements about the behaviour of salts dissolved in water is correct?

- A)** Ammonium chloride gives a basic solution, because the ammonium ion accepts a proton from water.
- B)** Sodium ethanoate gives a basic solution, because the ethanoate ion accepts a proton from water and releases OH^- .
- C)** Sodium ethanoate gives an acidic solution, because the sodium ion donates protons to water.
- D)** Sodium chloride gives a basic solution, because the chloride ion accepts protons from water.
- E)** Ammonium nitrate gives a basic solution, because the nitrate ion accepts protons from water.

Answer: B

The ethanoate ion is the conjugate base of a weak acid, so it accepts a proton from water, $\text{CH}_3\text{COO}^- + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{OH}^-$, releasing hydroxide and making sodium ethanoate basic. The ammonium claim reverses the chemistry, because NH_4^+ donates a proton to water and gives an acidic solution, and chloride and nitrate, as conjugate bases of strong acids, are far too weak to hydrolyse water.

39. A drip bag holds 500 cm^3 of glucose solution containing 9.0 g of glucose ($M = 180 \text{ g/mol}$); the label reads 1.8 % w/v, meaning 1.8 g of glucose per 100 cm^3 of solution. What is the molarity of the glucose?

- A)** 0.010 mol/dm^3 **B)** 0.050 mol/dm^3 **C)** 0.100 mol/dm^3 **D)** 0.180 mol/dm^3 **E)** 1.80 mol/dm^3

Answer: C

Step 1: $n(\text{glucose}) = 9.0 \div 180 = 0.050 \text{ mol}$

Step 2: the bag volume is $500 \text{ cm}^3 = 0.500 \text{ dm}^3$

Step 3: $c = 0.050 \div 0.500 = 0.10 \text{ mol/dm}^3$

This agrees with the label, since 1.8 g per 100 cm^3 gives 9.0 g in 500 cm^3 , and the 0.050 mol/dm^3 answer stops at the mole count without dividing by the bag's volume.

40. At 25°C the solubility of calcium hydroxide in water is 1.85 g per dm^3 of saturated solution, and the dissolved salt is fully dissociated. ($M(\text{Ca}(\text{OH})_2) = 74 \text{ g/mol}$; $K_w = 1.0 \times 10^{-14}$ at 25°C ; for dilute aqueous solutions 1 ppm is 1 mg per dm^3 .) Which of the following statements is correct?

- A)** The solubility is 0.025 mol/dm^3 and the pH of the saturated solution is 12.70.
- B)** The solubility is 0.050 mol/dm^3 and the pH of the saturated solution is 13.30.
- C)** The solubility is 0.050 mol/dm^3 and the pH of the saturated solution is 12.70.
- D)** The solubility is 0.025 mol/dm^3 and the calcium ion concentration is 1850 ppm.
- E)** The solubility is 0.025 mol/dm^3 and the pH of the saturated solution is 11.30.

Answer: A

Step 1: molar solubility = $1.85 \div 74 = 0.025 \text{ mol/dm}^3$

Step 2: full dissociation gives $[\text{OH}^-] = 2 \times 0.025 = 0.050 \text{ mol/dm}^3$

Step 3: $\text{pOH} = 2 - \log 5 = 1.30$, so $\text{pH} = 14 - 1.30 = 12.70$

The 0.050 mol/dm^3 figure reports the hydroxide concentration as the salt's solubility, and the calcium ion works out at $0.025 \times 40 = 1.0 \text{ g/dm}^3 = 1000 \text{ ppm}$, not 1850 ppm.

Section D - Redox and oxidation numbers (Q41-49)

41. What is the oxidation number of the carbon atom in methanal, HCHO ?

- A)** -2 **B)** 0 **C)** $+1$ **D)** $+2$ **E)** $+4$

Answer: B

Summing oxidation numbers over the neutral molecule HCHO : each H counts $+1$ toward carbon and O counts -2 , so $\text{C} + 2(+1) + (-2) = 0$, giving the aldehyde carbon an oxidation number of 0. For contrast, carbon sits at $+2$ in carbon monoxide and at $+4$ in carbon dioxide, so neither of those values belongs to an aldehyde carbon.

42. In the thiosulfate ion, $\text{S}_2\text{O}_3^{2-}$, the structure has a central sulfur atom bonded to three oxygen atoms and to one terminal sulfur atom, as if one oxygen of a sulfate unit had been replaced by sulfur. Which of the following statements about oxidation numbers in this ion is correct?

- A)** The average oxidation number of sulfur is +2, with the central sulfur at +5 and the terminal sulfur at -1.
- B)** The average oxidation number of sulfur is +4, because each sulfur behaves as in the sulfate ion.
- C)** Both sulfur atoms have oxidation number +2, because the charge of the ion is -2.
- D)** The terminal sulfur has oxidation number -2 and the central sulfur +4, exactly as in sulfate.
- E)** Oxidation numbers cannot be assigned, because the bond between identical atoms makes all the numbers undefined.

Answer: A

The ion's charge fixes only the sum: $2\text{S} + 3(-2) = -2$ gives an average of +2 for the two sulfur atoms. Structurally, the S-S bond contributes nothing because identical atoms share the bonding electrons equally, so the central sulfur bonded to three oxygens counts +5 and the terminal sulfur counts -1, and these sum to the required total of +4 for the two sulfurs. The numbers are fully assignable; a bond between identical atoms is ignored in the count, not undefined.

43. Which of the following statements about oxidation numbers in oxoanions is correct?

- A)** Sulfur is +7 in SO_4^{2-} and nitrogen is +4 in NO_2^- .
- B)** Sulfur is +6 in SO_4^{2-} and nitrogen is +5 in NO_2^- .
- C)** Sulfur is +4 in SO_4^{2-} and nitrogen is +3 in NO_2^- .
- D)** Sulfur is 0 in SO_4^{2-} because the oxygens carry the whole charge, and nitrogen is +3 in NO_2^- .
- E)** Sulfur is +6 in SO_4^{2-} and nitrogen is +3 in NO_2^- .

Answer: E

In SO_4^{2-} , $\text{S} + 4(-2) = -2$ gives $\text{S} = +6$; in NO_2^- , $\text{N} + 2(-2) = -1$ gives $\text{N} = +3$

The +4 value for sulfur belongs to sulfite, SO_3^{2-} , and +5 for nitrogen belongs to nitrate, NO_3^- , so any statement using those numbers describes the wrong oxoanion.

44. Ethanol, $\text{CH}_3\text{CH}_2\text{OH}$, is oxidised to ethanal, CH_3CHO . Which of the following statements is correct?

- A)** The carbon bearing the OH group goes from oxidation number -1 to +1, so ethanol is oxidised.
- B)** Both carbon atoms rise from -3 to +1, so ethanol is oxidised.
- C)** The oxidation number of oxygen rises from -2 to 0, so oxygen is oxidised.
- D)** The average oxidation number of carbon is unchanged, so no redox change occurs.
- E)** The carbon bearing the OH group goes from oxidation number +1 to -1, so ethanol is reduced.

Answer: A

In ethanol, the CH_2OH carbon carries two C-H bonds (each counts -1 for carbon) and one C-O bond (+1), giving an oxidation number of -1. In ethanal, the same carbon carries one C-H (-1) and a C=O double bond (+2), giving +1. The rise from -1 to +1 is a loss of two electrons, so ethanol is oxidised. The claim that the carbon falls from +1 to -1 reverses the direction and would describe reduction, and the average over both carbons also moves, from -2 to -1, so no unchanged-average reading survives.

45. In aqueous solution, $2\text{FeCl}_3 + \text{SnCl}_2 \rightarrow 2\text{FeCl}_2 + \text{SnCl}_4$. Which species acts as the oxidising agent?
- A) FeCl_2 , because iron ends at oxidation number +2.
 - B) FeCl_3 , because Fe^{3+} is reduced to Fe^{2+} .
 - C) Cl^- , because it keeps oxidation number -1 in every compound.
 - D) SnCl_2 , because tin is reduced from +2 to +4.
 - E) SnCl_4 , because tin is oxidised from +2 to +4.

Answer: B

Tin rises from +2 in SnCl_2 to +4 in SnCl_4 , so it is oxidised and SnCl_2 is the reducing agent. Iron falls from +3 in FeCl_3 to +2 in FeCl_2 , so Fe^{3+} gains electrons, and the species that is reduced is by definition the oxidising agent, which is FeCl_3 . Chloride stays at -1 throughout and is a spectator, and an oxidation product such as SnCl_4 cannot itself be the oxidising agent.

46. In the laboratory preparation of chlorine, MnO_2 reacts with concentrated hydrochloric acid: $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$. Which of the following statements is correct?
- A) HCl is the oxidising agent, because it supplies the hydrogen ions.
 - B) MnO_2 is the reducing agent, because manganese goes from +4 to +2.
 - C) Chlorine gas is the reducing agent, because it is the oxidation product.
 - D) No oxidation occurs, because no oxygen is transferred between the reactants.
 - E) MnO_2 is the oxidising agent and the chloride ion is the reducing agent.

Answer: E

Manganese falls from +4 in MnO_2 to +2 in MnCl_2 , so MnO_2 is reduced and therefore acts as the oxidising agent, while chloride rises from -1 in HCl to 0 in Cl_2 and is oxidised, making the chloride ion the reducing agent. Calling MnO_2 the reducing agent reverses the rule that the reduced species is the oxidising one, and redox concerns electron transfer, so the absence of transferred oxygen does not mean no oxidation occurs.

47. Chlorine gas is bubbled into cold dilute sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$. Which of the following statements is correct?
- A) Chlorine is oxidised only, and the hydroxide ion is the oxidising agent.
 - B) Chlorine is both oxidised and reduced, while oxygen keeps oxidation number -2 throughout.
 - C) The chloride ion is oxidised, because chlorine gas gains electrons.
 - D) Oxygen is oxidised from -2 to 0 in the water that forms.
 - E) Chlorine is reduced only, and the hydroxide ion is the reducing agent.

Answer: B

Chlorine starts at oxidation number 0 and finishes in two states, -1 in NaCl (reduced) and $+1$ in NaClO (oxidised), so chlorine is both oxidised and reduced: a disproportionation. Oxygen remains at -2 in OH^- , ClO^- and H_2O throughout, and the hydroxide ion changes no oxidation state, so it is neither agent. Note also that gaining electrons is reduction, so any claim that a species is oxidised because it gains electrons contradicts itself.

48. Sulfite ions, SO_3^{2-} , are oxidised to sulfate ions, SO_4^{2-} , in acidic solution. When this change is written as a balanced half-equation, how many electrons appear on the right-hand side?
- A) 1 B) 2 C) 3 D) 4 E) 6

Answer: B

Sulfur rises from +4 in SO_3^{2-} to +6 in SO_4^{2-} , a loss of two electrons, so the balanced half-equation in acid is $\text{SO}_3^{2-} + \text{H}_2\text{O} \rightarrow \text{SO}_4^{2-} + 2\text{H}^+ + 2\text{e}^-$, and the charge check agrees: -2 on the left, and $-2 + 2 - 2 = -2$ on the right. The electron count follows the two-unit change of the sulfur, not the number of oxygen atoms involved.

49. In acidic solution, manganate(VII) ions, MnO_4^- , oxidise iron(II) ions, Fe^{2+} , to Fe^{3+} , the manganese ending as Mn^{2+} . In the balanced ionic equation, how many electrons are transferred per MnO_4^- ion, and what coefficient does Fe^{2+} carry?

- A)** 3 electrons and 3 Fe^{2+}
- B)** 4 electrons and 4 Fe^{2+}
- C)** 5 electrons and 1 Fe^{2+}
- D)** 5 electrons and 5 Fe^{2+}
- E)** 7 electrons and 7 Fe^{2+}

Answer: D

Step 1: manganese falls from +7 in MnO_4^- to +2 in Mn^{2+} , so each permanganate ion gains 5 electrons.

Step 2: each Fe^{2+} loses 1 electron to become Fe^{3+} , so five Fe^{2+} ions are required, giving $\text{MnO}_4^- + 8\text{H}^+ + 5\text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} + 5\text{Fe}^{3+}$, with charge and atoms balanced. The 7-electron answer wrongly counts down to oxidation number 0, the metallic state.

Section E - Bonding, polarity and intermolecular forces (Q50-59)

50. Butane (C_4H_{10} , Mr 58), propanone ($\text{C}_3\text{H}_6\text{O}$, Mr 58) and propan-1-ol ($\text{C}_3\text{H}_8\text{O}$, Mr 60) boil at approximately 0 °C, 56 °C and 97 °C, in some order. Which assignment is correct?

- A)** Butane 0 °C, propan-1-ol 56 °C, propanone 97 °C.
- B)** Butane 56 °C, propanone 97 °C, propan-1-ol 0 °C.
- C)** Butane 0 °C, propanone 56 °C, propan-1-ol 97 °C.
- D)** Propanone 0 °C, butane 56 °C, propan-1-ol 97 °C.
- E)** Propan-1-ol 0 °C, propanone 56 °C, butane 97 °C.

Answer: C

With molar masses nearly equal (58, 58, 60), boiling point is decided by the strength of intermolecular forces. Propan-1-ol hydrogen bonds through its O-H group and boils highest at 97 °C, propanone has permanent dipole-dipole forces only and boils at 56 °C, and butane has only weak London forces and boils lowest at 0 °C. Any ranking that places propanone above propan-1-ol fails because dipole-dipole attraction is weaker than hydrogen bonding at the same molar mass.

51. Butan-1-ol, butan-2-ol and 2-methylpropan-2-ol are isomers that all contain an O-H group. Their boiling points are approximately 118 °C, 100 °C and 82 °C respectively. Which statement correctly explains this trend?

- A)** Branching lowers the boiling point because it weakens the O-H bonds inside the molecules.
- B)** All three have the same boiling point, because hydrogen bonding dominates in each of them.
- C)** The boiling points differ because the three isomers contain different functional groups.
- D)** 2-methylpropan-2-ol has the highest boiling point, because its compact shape packs the molecules closest together.
- E)** Butan-1-ol has the highest boiling point, because its least branched chain gives the greatest molecular surface contact and so the strongest London forces, which act in addition to the hydrogen bonding present in all three isomers.

Answer: E

All three isomers hydrogen bond equally through their O-H groups, so the boiling-point spread comes from the London force component, which grows with molecular surface contact. The unbranched chain of butan-1-ol gives the greatest contact and the highest boiling point (118 °C), while the compact branching of 2-methylpropan-2-ol gives the least contact and the lowest (82 °C). Branching alters intermolecular contact only; it cannot weaken the O-H covalent bonds inside the molecules, and all three carry the same functional group.

52. Hydrogen fluoride (Mr 20) boils at approximately 20 °C, while hydrogen chloride (Mr 36.5) boils at approximately -85 °C. Which statement explains this comparison?

- A)** HF molecules hydrogen bond to one another, a force much stronger than the dipole-dipole and London forces between HCl molecules, and this outweighs the higher molar mass of HCl.
- B)** HCl has the higher molar mass, so its London forces should make it boil higher, and the quoted temperatures cannot both be right.
- C)** The H-F covalent bond is stronger than the H-Cl covalent bond, and covalent bond strength sets the boiling point.
- D)** HF is ionic while HCl is covalent, and ionic compounds boil much higher.
- E)** HCl is non-polar, so only very weak London forces act between its molecules, while HF is also non-polar but lighter.

Answer: A

Fluorine's high electronegativity lets HF molecules form strong hydrogen bonds, while HCl molecules attract each other only by ordinary dipole-dipole and London forces, so HF boils far higher (about 20 °C versus -85 °C) despite its lower molar mass. The comparison concerns intermolecular forces, not covalent bond strength, because the H-F bonds survive boiling intact; and HCl is polar, just with a weaker dipole and no hydrogen bonding.

53. Which of the following statements is correct?

- A)** Water is non-polar, because its two identical O-H bonds pull equally in opposite directions.
- B)** Carbon dioxide is polar, because each C=O bond is strongly polar.
- C)** Boron trifluoride is non-polar, because its three identical B-F bond dipoles cancel in the trigonal planar arrangement.
- D)** Ammonia is non-polar, because the lone pair on nitrogen makes the molecule symmetrical.
- E)** Any molecule that contains polar bonds must be polar overall.

Answer: C

BF₃ is trigonal planar, and its three identical B-F dipoles at 120 degrees cancel vectorially, so the molecule is non-polar. Water is bent, so its two O-H dipoles leave a resultant and the molecule is polar; CO₂ is linear and cancels, and ammonia is pyramidal and polar. The general claim that polar bonds always make a polar molecule is disproved by every symmetric case such as BF₃ and CO₂.

54. Carbon tetrachloride, CCl₄, chloroform, CHCl₃, and dichloromethane, CH₂Cl₂, are all tetrahedral molecules. Which statement about their polarity is correct?

- A)** All three are polar, because every tetrahedral molecule containing polar bonds is polar overall.
- B)** The polarity of each molecule is decided only by its molar mass, so the heaviest molecule is the most polar.
- C)** CCl₄ is the most polar of the three, because it contains the greatest number of C-Cl dipoles.
- D)** All three are non-polar, because the tetrahedral shape of each molecule cancels every bond dipole.
- E)** CCl₄ is non-polar because its four identical C-Cl dipoles cancel, while CHCl₃ and CH₂Cl₂ are polar because the C-H bonds do not balance the C-Cl dipoles.

Answer: E

In CCl₄ the four identical C-Cl dipoles sit symmetrically in the tetrahedron and cancel exactly, so it is non-polar. Substituting Cl by H destroys the symmetry of the dipole arrangement, so CHCl₃ and CH₂Cl₂ both keep a net dipole and are polar. Adding more identical dipoles in a fully symmetric tetrahedron makes the resultant smaller, not larger, so CCl₄ is the least polar of the three, not the most.

55. Carbon dioxide and sulfur dioxide both contain polar bonds to oxygen, yet CO₂ is non-polar and SO₂ is polar. Why?

- A)** CO₂ is linear, so its two C=O bond dipoles point in opposite directions and cancel, while SO₂ is bent, so its two S=O dipoles add to give a net dipole.
- B)** Both molecules are linear, but only SO₂ has a lone pair on the central atom strong enough to create a dipole.
- C)** SO₂ is bent, which makes its S=O bonds polar, while the linear shape of CO₂ makes its C=O bonds non-polar.
- D)** SO₂ is polar because its molecules are heavier than CO₂ molecules and attract each other more strongly.
- E)** The C=O bonds in CO₂ are non-polar, because carbon and oxygen have almost equal electronegativities.

Answer: A

Both molecules contain polar bonds, so geometry decides the outcome. CO₂ is linear, its two equal C=O dipoles point in opposite directions and cancel; SO₂ is bent (the lone pair on sulfur holds the O-S-O angle near 119 degrees), so its two S=O dipoles add to a net dipole. Bond polarity depends on electronegativity difference, not on molecular shape, so the claim that bending changes whether the bonds themselves are polar is wrong.

56. Four unknown substances were tested, with these results. P: melting point 801 °C, conducts electricity only when molten or dissolved in water. Q: melting point –95 °C, never conducts electricity. R: melting point 1085 °C, conducts electricity as a solid. S: melting point 1710 °C, never conducts electricity, even when molten. Which statement correctly identifies one of the substances?

- A)** R is metallic, because conduction in the solid state requires mobile delocalised electrons.
- B)** Q is ionic, because ionic bonds are weak and break at a low temperature.
- C)** S is ionic, because its very high melting point proves strong ionic bonds.
- D)** P is metallic, because it conducts electricity when molten.
- E)** P is a simple molecular substance, because molecular substances dissolve well in water.

Answer: A

Conduction through a solid requires mobile charge carriers, and only the delocalised electrons of a metal can move through a rigid lattice, so R, which conducts as a solid, is metallic. P conducts only when molten or dissolved, the signature of an ionic solid whose ions are locked until melting or solution frees them, and S, which never conducts even when molten, is giant covalent rather than ionic, because molten ionic compounds do conduct, so a very high melting point alone cannot mark ionic bonding.

57. Sodium chloride melts at 801 °C while iodine melts at 114 °C, although iodine has the larger molar mass. Which statement explains the comparison?

- A)** Iodine melts easily because the covalent bonds between its molecules are weak.
- B)** Sodium chloride has the higher melting point because its molar mass is greater than that of iodine.
- C)** Melting sodium chloride means overcoming strong electrostatic attractions between oppositely charged ions in a giant lattice, while melting iodine only overcomes weak London forces between I₂ molecules, and the I-I covalent bonds stay intact.
- D)** Sodium chloride contains covalent bonds, which are stronger than the covalent bonds inside iodine molecules.
- E)** Both solids are giant lattices, and iodine simply has a lattice built from lighter particles.

Answer: C

Melting NaCl means overcoming the strong electrostatic attractions between Na⁺ and Cl[–] ions repeated throughout a giant lattice, which takes 801 °C. Solid iodine is simple molecular, so melting only overcomes the weak London forces between I₂ molecules while every I-I covalent bond stays intact, which takes only 114 °C. The forces between separate iodine molecules are intermolecular London forces, not covalent bonds, and molar mass cannot compare an ionic lattice with a molecular solid.

58. Use these average bond energies: H-H 436 kJ/mol, Cl-Cl 242 kJ/mol, H-Cl 431 kJ/mol. For the reaction $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$, the enthalpy change is approximately:

- A)** -368 kJ/mol **B)** -247 kJ/mol **C)** -184 kJ/mol **D)** +184 kJ/mol **E)** +247 kJ/mol

Answer: C

Step 1: bonds broken cost $436 (\text{H-H}) + 242 (\text{Cl-Cl}) = 678 \text{ kJ/mol}$

Step 2: bonds formed release $2 \times 431 = 862 \text{ kJ/mol}$

Step 3: $\Delta H = 678 - 862 = -184 \text{ kJ/mol}$, negative because stronger bonds form than break. Reversing the subtraction produces +184, and the 247 magnitudes forget that two H-Cl bonds form per equation as written.

59. The thermal decomposition of calcium carbonate, $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$, has $\Delta H = +178 \text{ kJ/mol}$. Which statement is correct?

- A)** Bonds formed in the products are stronger than bonds broken in the reactants, which explains the positive ΔH .
B) The products are lower in enthalpy than the reactants, so heat is released to the surroundings.
C) At a sufficiently high temperature ΔH becomes negative, so the decomposition turns exothermic.
D) The positive sign means energy is released by the system, so the reaction is exothermic.
E) The products are higher in enthalpy than the reactants, and the difference is absorbed as heat from the surroundings.

Answer: E

A positive ΔH marks an endothermic change: the products sit higher in enthalpy than the reactants and the difference is absorbed as heat from the surroundings. The claim that stronger bonds form in the products is backwards, because stronger bonds forming releases energy and would drive ΔH negative, and ΔH does not flip sign merely because the mixture is hot.

Section F - Kinetics and catalysis (Q60-65)

60. For reactions in dilute solution at a fixed temperature, the units of reaction rate are always $\text{mol}/(\text{dm}^3 \cdot \text{s})$. Which statement about rate constants is correct?

- A)** A zero-order rate constant has the units s^{-1} .
B) A first-order rate constant has the units s^{-1} , and a second-order rate constant has the units $\text{dm}^3/(\text{mol} \cdot \text{s})$.
C) A first-order rate constant has the units $\text{mol}/(\text{dm}^3 \cdot \text{s})$.
D) Every rate constant has the same units as the rate itself, $\text{mol}/(\text{dm}^3 \cdot \text{s})$, whatever the order of reaction.
E) A second-order rate constant has the units $\text{mol}/(\text{dm}^3 \cdot \text{s})$.

Answer: B

In $\text{rate} = k[\text{A}]^n$, k must carry whatever units make the right-hand side come out in $\text{mol}/(\text{dm}^3 \cdot \text{s})$. Zero order gives k the rate's own units, first order gives s^{-1} , and second order gives $\text{dm}^3/(\text{mol} \cdot \text{s})$. So the correct pairing is a first-order constant in s^{-1} with a second-order constant in $\text{dm}^3/(\text{mol} \cdot \text{s})$; attaching $\text{mol}/(\text{dm}^3 \cdot \text{s})$ to a first-order constant actually describes the zero-order case.

61. A drug is removed from the bloodstream by first-order kinetics with a half-life of 4 hours. Starting from an initial dose D , the amount still present 12 hours after administration is:

- A)** 1/16 of D **B)** 1/8 of D **C)** 1/6 of D **D)** 1/4 of D **E)** 1/2 of D

Answer: B

Twelve hours contains $12 \div 4 = 3$ half-lives, and each first-order half-life halves whatever amount is present: $D \rightarrow D/2 \rightarrow D/4 \rightarrow D/8$. One eighth of the dose remains. Dividing one half by three to get 1/6 treats the decay as linear when it is exponential.

62. For the reaction $A + B \rightarrow \text{products}$, kept at constant temperature: doubling $[A]$ while holding $[B]$ constant doubles the rate, and doubling $[B]$ while holding $[A]$ constant increases the rate fourfold. Which statement is correct?

- A)** The reaction is second order overall, and k has the units $\text{dm}^3/(\text{mol}\cdot\text{s})$.
- B)** The rate equation is $\text{rate} = k[A][B]^2$, the overall order is three, and k has the units $\text{dm}^3/(\text{mol}\cdot\text{s})$.
- C)** The reaction is fourth order overall, because the fourfold increase in the second experiment means the exponent of B is three.
- D)** The rate equation is $\text{rate} = k[A][B]^2$, the overall order is three, and k has the units $\text{dm}^6/(\text{mol}^2\cdot\text{s})$.
- E)** The reaction is first order overall, and k has the units s^{-1} , because each reactant contributes one power.

Answer: D

Step 1: doubling $[A]$ doubling the rate fixes A 's exponent at 1, and doubling $[B]$ quadrupling the rate fixes B 's exponent at 2, so $\text{rate} = k[A][B]^2$ and the overall order is $1 + 2 = 3$.

Step 2: $k = \text{rate} \div ([A][B]^2)$, so the units are $(\text{mol}/(\text{dm}^3\cdot\text{s})) \div (\text{mol}/\text{dm}^3)^3 = \text{dm}^6/(\text{mol}^2\cdot\text{s})$

The $\text{dm}^3/(\text{mol}\cdot\text{s})$ units belong to a second-order constant and cannot serve an overall third-order rate law.

63. The Haber process, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ with $\Delta H = -92 \text{ kJ/mol}$, uses an iron catalyst. Which statement is correct?

- A)** The catalyst lowers the activation energy of the forward and reverse reactions equally, so equilibrium is reached sooner, but the equilibrium position and ΔH are unchanged.
- B)** The catalyst raises the activation energy of the reverse reaction, so ammonia accumulates.
- C)** The catalyst works by raising the average kinetic energy of the gas molecules, so that more collisions are effective.
- D)** The catalyst makes the reaction more exothermic, which increases the yield at the same temperature.
- E)** The catalyst shifts the equilibrium position to the right, increasing the final yield of ammonia.

Answer: A

A catalyst offers a lower-activation-energy pathway and speeds the forward and reverse reactions by the same factor, so the mixture reaches equilibrium sooner while the equilibrium position, K and ΔH all stay unchanged. The claim that it shifts equilibrium toward ammonia fails because the yield is fixed by thermodynamics, and raising the average kinetic energy of molecules is what heating does, not what a catalyst does.

64. Manganese(IV) oxide speeds up the decomposition of hydrogen peroxide, $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$, and the black powder can be recovered unchanged at the end. The best explanation is that MnO_2 :

- A)** is consumed to supply the energy that the reaction needs, which is why a little disappears each time.
- B)** raises the average kinetic energy of the H_2O_2 molecules, so more collisions exceed the activation energy.
- C)** shifts the equilibrium of the decomposition towards the products, so the reaction goes faster.
- D)** reduces the enthalpy change of the decomposition, which releases more energy to drive the reaction.
- E)** provides an alternative reaction pathway of lower activation energy and is regenerated during the cycle, so it is not consumed.

Answer: E

MnO_2 provides the decomposition with an alternative pathway of lower activation energy and is regenerated as the cycle turns, which is exactly why the same black powder can be recovered unchanged and reused. The solid is not consumed (its recovery disproves that), a catalyst does not alter the enthalpy change of the reaction, and raising molecular speeds is what heating accomplishes, not catalysis.

65. A reaction mixture is heated from 25 °C to 55 °C and the rate rises sharply. According to collision theory, the main reason is that:

- A)** molecules collide much more often, and this increase in collision frequency alone explains the whole rate rise.
- B)** faster molecules are more likely to meet in the correct orientation, because high speed corrects the approach angle.
- C)** the activation energy of the reaction decreases as the temperature rises.
- D)** a much larger fraction of molecules now has energy at or above the activation energy, so far more collisions are effective.
- E)** the enthalpy change of the reaction becomes more negative at the higher temperature.

Answer: D

Heating reshapes the Maxwell-Boltzmann energy distribution so that a much larger fraction of molecules has energy at or above the activation energy, and it is this surge in effective collisions that drives the sharp rate rise. The activation energy is a property of the reaction pathway and does not fall when the mixture is warmed, and collision frequency rises only slightly over a 30 degree change, far too little to account for the whole effect.